

The BDPP Theorem for Projective Manifolds

Kähler MMP

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This note has two aims:

- (1) We will study the rational curves on Kähler varieties ([?], [?]),
- (2) We will give an introduction to the BDPP theorem for projective and Kähler manifolds ([?], [?]).

Some further discussion and applications can be found in [Cone Theorem for Kähler MMP](#).

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1 Duality Between Various Positive Cones

In this section, we study the duality between various cones in the projective and Kähler settings.

The following theorem shows the duality between the pseudo-effective cone and the movable cone on a projective manifold.

Lemma 1.1. *Let X be a projective manifold. Let $\gamma \in N_1(X)$ be a movable class. Given any prime divisor E , there exists a representative γ_E such that γ_E intersects E properly and $\gamma \equiv \gamma_E$.*

Remark 1.2. It is unclear whether this result also holds for Kähler manifolds.

Proof. ■

Theorem 1.3. *Let X be a projective manifold. Then the pseudo-effective cone is dual to the cone of movable curves:*

$$\mathcal{E} = \overline{\text{Mov}}(X)^\vee.$$

In other words, a divisor is pseudo-effective if and only if it has non-negative intersection with every movable curve.

Remark 1.4. David [?] proved ...

Remark 1.5. Let us briefly sketch the idea of the proof.

Proof. Let C be a movable curve. By Lemma 1.1, we can choose some C' such that $C \equiv C'$ and C' meets the given pseudo-effective divisor properly. Hence

$$\mathcal{E} \subset \overline{\text{Mov}}(X)^\vee.$$

Conversely, if the inclusion is strict, then by the strict separation theorem for closed convex cones, there exists

$$\xi \in \partial(\mathcal{E}(X)), \quad \xi \in \text{int}(\overline{\text{Mov}}(X)^*).$$

We want to deduce a contradiction. Since X is projective, we can find an ample divisor H such that $\xi - \epsilon H$ still lies in the movable cone. So that

$$\frac{(\xi \cdot C)}{(H \cdot C)} \geq \epsilon, \quad \forall C \in \overline{\text{Mov}}(X).$$

On the other hand, we can apply Fujita approximation to the class $\xi + tH$ for the ample H , and obtain a modification

$$\mu_t: X_t \rightarrow X$$

such that $\mu_t^*(\xi + tH) = A_t + E_t$. Choose $C = (\mu_t)_* A_t^{n-1}$, then apply the asymptotic orthogonality of Fujita approximation to $\xi \cdot C$ and the Teissier–Hovanskii inequality to deduce an upper bound

$$\delta_t \geq \frac{\xi \cdot C}{H \cdot C} \geq \epsilon$$

with $\delta_t \rightarrow 0$ when $t \rightarrow 0$ (here δ_t is a constant depending on the volume of A_t ; since $\text{vol}(\xi) = 0$, by Fujita approximation $\text{vol}(A_t) \rightarrow 0$ when $t \rightarrow 0$). This contradicts $\delta_t \rightarrow 0$, completing the proof. ■

One can generalize the duality theorem to normal Moishezon spaces using a standard blow-up argument.

Theorem 1.6. *Let X be a normal Moishezon space. Then the pseudo-effective cone is dual to the movable cone of curves.*

Proof. ■

Using the duality theorem, we can show that the cone of nef curves coincides with the movable cone of curves.

Theorem 1.7. *Let X be a normal Moishezon space. Then the Batyrev nef cone coincides with the movable cone of curves.*

2 BDPP Theorem for Projective Manifolds

The projective uniruled manifold is characterized by the pseudo-effectiveness of the canonical bundle.

Lemma 2.1. *Given a movable curve C , there exists a covering family $\bigcup_{t \in S} C_t$ containing C , which covers a dense open subset of X . To be more precise, we can find a diagram*

$$\begin{array}{ccc} C & \xrightarrow{\phi} & X \\ f \downarrow & & \\ S & & \end{array}$$

where f is a fibration with fibers C_t , ϕ is a dominant generically finite morphism, and $\{C_t\}_{t \in S}$ lie in the same numerical class as C .

Proof. ■

Theorem 2.2 ([?, Corollary 0.3]). *Let X be a projective manifold. Then X is uniruled if and only if K_X is not pseudo-effective.*

Remark 2.3. One direction of the proof is easy, and can be adapted to Kähler manifolds. The converse direction — that K_X not being pseudo-effective implies X is uniruled — is non-trivial, and requires the Mori bend-and-break technique and the duality between the pseudo-effective cone and the movable cone.

Remark 2.4. Miyaoka and Mori [?] proved that a projective manifold is uniruled if and only if there exists an open subset over which there passes a K_X -negative curve. For more discussion about the Miyaoka–Mori theorem (and various properties of uniruled manifolds) see Note 15.

Proof. It suffices to prove that if K_X is not pseudo-effective, then X is uniruled. By duality of the pseudo-effective cone and the movable cone, we know that there exists a movable curve C such that

$$K_X \cdot C < 0.$$

By Lemma 2.1, we can produce a covering family of K_X -negative irreducible curves using the movable curve C . ■

We can generalize the BDPP theorem to the singular case.

Theorem 2.5. *Let (X, B) be a $\mathbb{Q}$ -factorial log pair. If $K_X + B$ is not pseudo-effective, then X is uniruled.*

Remark 2.6. Rational curves on singular spaces are subtle. For further discussion, see Note 9: Rational Curves on Moishezon Spaces and Kähler Varieties.

Proof. Take the log resolution

$$f: X' \rightarrow X,$$

such that $f^*(K_X + B) = K_{X'} + B'$. Since being uniruled is a birational invariant, if X is not uniruled, then neither is X' . Then by the BDPP theorem proved above, $K_{X'}$ is pseudo-effective, thus K_X is pseudo-effective. Since B is effective, $K_X + B$ is also pseudo-effective. ■

We can also show the converse direction for canonical singularities.

Theorem 2.7.

However, the following example indicates that the BDPP theorem may fail for singular varieties.

We can characterize uniruled varieties using subsheaves of the tangent sheaf.

Theorem 2.8. *Let X be a projective manifold and let $\mathcal{F} \subset T_X$ be a coherent subsheaf of rank r . If $\det \mathcal{F}^*$ is not pseudo-effective, then X is uniruled.*

Proof. ■

3 Several Applications of BDPP Theorem

3.1 Applications of duality of pseudo-effective cone and cone of movable curves

3.2 Producing rational curves using BDPP conjecture

3.3 Cone theorem using BDPP conjecture

[?] proved the following cone theorem for klt Kähler pairs. We will discuss the details of the proof in [REFERENCE].

Theorem 3.1.